

BASHAR ALMUNTASER

AI Engineer | Machine Learning Engineer



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Professional Summary

AI Engineer with hands-on experience building and deploying production-grade machine learning systems, LLM pipelines, NLP models, and computer vision applications. Skilled in end-to-end ML engineering — from fine-tuning transformer models (BERT, RoBERTa) to deploying containerized microservices on Linux servers. Experienced in multi-model LLM orchestration, bilingual Arabic/English NLP, AI agent development using LangChain, and real-time data pipelines. Strong focus on Arabic and Gulf market AI applications.

Education

2022 – 2026
Sana'a, Yemen

Bachelor of Science in Artificial Intelligence
Emirates International University

Skills

Programming Languages

Python, TypeScript, SQL, Java

Natural Language Processing

Arabic NLP, CAMEL-BERT, BiLSTM, pyarabic, Unicode Normalization, Named Entity Recognition (NER), Sentiment Analysis, Text Classification, Domain-Adversarial Training

Data Engineering

PySpark, Apache Kafka, Pandas, NumPy, TimescaleDB, Elasticsearch

Frontend Development

React, Next.js, TypeScript

DevOps & Infrastructure

Docker, Docker Compose, Linux Server Administration, Git, GitHub, Langfuse, Prisma ORM

Machine Learning & AI

PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, BERT, RoBERTa, Fine-tuning, Transfer Learning, RAG (Retrieval-Augmented Generation), LLM APIs (OpenAI GPT-4, Anthropic Claude, Google Gemini, Perplexity), LangChain, AI Agents, Prompt Engineering

Computer Vision

OpenCV, MediaPipe, Facial Landmark Detection, Eye Aspect Ratio (EAR), 3D Pose Estimation, Real-Time Inference

Backend Development

FastAPI, NestJS, REST APIs, Celery, BullMQ, Python Workers

Databases

PostgreSQL, MongoDB, Redis

Languages

Arabic

Native

English

Professional Working Proficiency

Projects

2025 – 2026

AI Discovery Monitor — Generative Engine Optimization (GEO) Platform

Solo ML Engineer | Stack: Python, NestJS, Next.js, PostgreSQL, Redis, Docker, OpenAI API, Claude API, Gemini API, Perplexity API, LangChain, Langfuse

- Designed and implemented an 8-stage asynchronous AI analysis pipeline executing structured prompt workflows across GPT-4, Claude, Gemini, and Perplexity using a Python worker architecture with BullMQ/Redis task queues and Pydantic schema validation.

- Built a bilingual Arabic/English entity resolution engine using Unicode normalization, ALA-LC transliteration, legal suffix stripping, Jaro-Winkler fuzzy matching, and PostgreSQL trigram (pg_trgm) indexing — achieving 70–80% accuracy on Arabic-only Gulf market entities.
- Engineered a hallucination detection and claim verification layer that cross-references LLM-extracted brand claims against live web content, Wikidata SPARQL endpoints, and Google Knowledge Graph API using multi-signal confidence scoring.
- Designed a data-driven recommendation engine performing competitive feature gap analysis across Wikipedia presence, Schema.org markup, Wikidata entries, and press citation signals — scoring recommendations by Spearman rank correlation against brand mention frequency.
- Architected production observability infrastructure including SHA-256 prompt version control, model drift detection via canary entity monitoring, LLM cost and latency tracking with Langfuse, and statistical significance enforcement using McNemar's test (minimum 50 runs per entity per model).

2025 – 2026

SAPA — Smart Amazon Product Analyzer

Solo Full-Stack ML Engineer | Stack: Python, FastAPI, React, TypeScript, PostgreSQL, TimescaleDB, Elasticsearch, Redis, Celery, Docker, LightGBM, BERT, LLaMA-3, Ollama

- Engineered a five-indicator product scoring engine combining LightGBM demand forecasting with a hybrid BERT and LLaMA-3 NLP pipeline (via Ollama) for review toxicity detection and competitive market analysis using the Herfindahl-Hirschman Index (HHI).
- Designed a Margin Kill-Switch automation that rejects investment opportunities when net ROI falls below configurable thresholds based on real-time Amazon FBA fee calculations.
- Deployed an 8-container Docker Compose microservices architecture (FastAPI, React 19, TimescaleDB, Elasticsearch, Celery, Redis, Ollama, Nginx) on Linux servers with health-check-ordered startup sequencing.
- Built an executive analytics dashboard with radar charts and time-series visualizations classifying products as RECOMMENDED, MONITOR, or REJECT based on institutional investment thresholds.

2024 – 2025

Real-Time Driver Drowsiness Detection System

Solo ML Engineer | Stack: Python, OpenCV, MediaPipe, SpeechRecognition, pytsx3, Multithreading

- Built a real-time drowsiness detection pipeline using OpenCV and MediaPipe FaceMesh to analyze Eye Aspect Ratio (EAR) and facial landmark geometry for driver alertness classification.
- Implemented head posture monitoring through 3D pose estimation and angular deviation analysis to detect fatigue-related positional drift before critical impairment occurs.
- Developed an AI-powered voice assistant using SpeechRecognition and pytsx3 for hands-free driver interaction and proactive safety alerts.
- Architected a multi-threaded event-driven system handling concurrent video processing, audio notifications, and voice responses without blocking the core inference loop — with temporal state tracking to reduce false alarms.

2024 – 2024

Financial Fraud Detection Platform

ML Engineer | Stack: Python, PySpark, Apache Kafka, MongoDB, Scikit-learn

- Built an end-to-end fraud detection pipeline using PySpark and ensemble machine learning models to classify anomalous financial transactions at scale.
- Simulated real-time transaction streaming using Apache Kafka, engineering fraud-signal features and storing model predictions in MongoDB for downstream tracking and analysis.

2024 – 2024

Arabic Sentiment Analysis and Fake Review Detection

ML Engineer | Stack: Python, CAMEL-BERT, BiLSTM, Hugging Face Transformers, FastAPI, Domain-Adversarial Training

- Fine-tuned CAMEL-BERT and BiLSTM architectures for domain-specific Arabic sentiment classification, building custom Arabic text preprocessing pipelines to handle social media dialects and cross-domain generalization.
- Developed a domain-adversarial transformer-based NLP system for cross-category fake review detection, deployed as a real-time inference service using FastAPI.